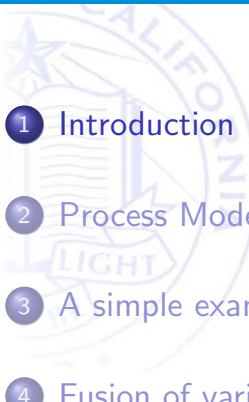


The Kalman Filter

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Outline

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- 1 Introduction
 - 2 Process Model
 - 3 A simple example
 - 4 Fusion of variables
 - 5 Discrete Kalman Filter
 - 6 Extended Kalman Filter
 - 7 Summary

Introduction

- Recapitulation of system models
- Integration of stochastic variables
- How to perform fusion in a more general sense
- Doing this in a non-linear system

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State space model



$$\begin{aligned}s_t &= Fs_{t-1} + Gu_t + w_t \\ z_t &= Hs_t + v_t\end{aligned}$$

- where F is the system model, G is the deterministic input, H is a prediction of where features are in the world, w is the system noise, and v is the measurement noise
- $p(w) \sim N(0, Q)$
- $p(v) \sim N(0, R)$

State state model example

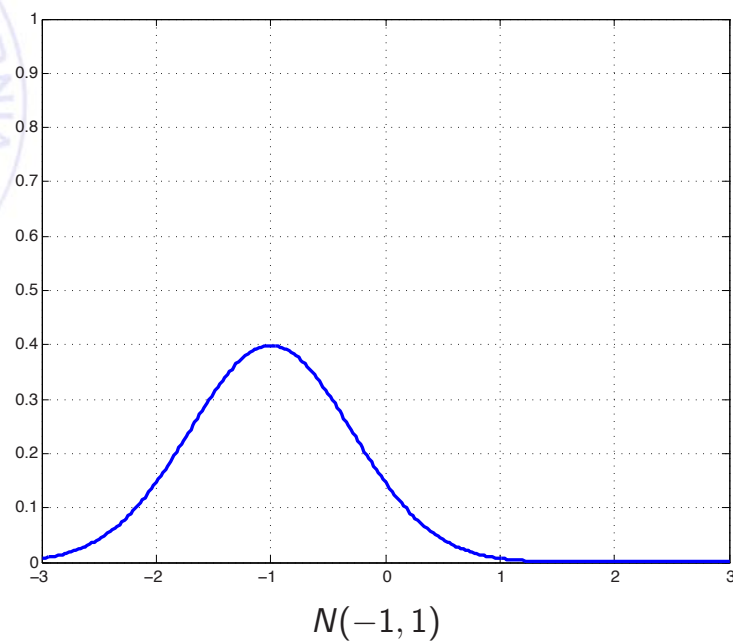


$$\begin{aligned}x_t &= x_{t-1} + v_{t-1}T + \frac{1}{2}a_{t-1}T^2 \\ v_t &= v_{t-1} + a_{t-1}T \\ a_t &= a_{t-1} \\ s_t &= \begin{bmatrix} x_t \\ v_t \\ a_t \end{bmatrix} \\ \mathbf{F} &= \begin{bmatrix} 1 & T & \frac{1}{2}T^2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix} \\ \mathbf{G} &= [001]^T\end{aligned}$$

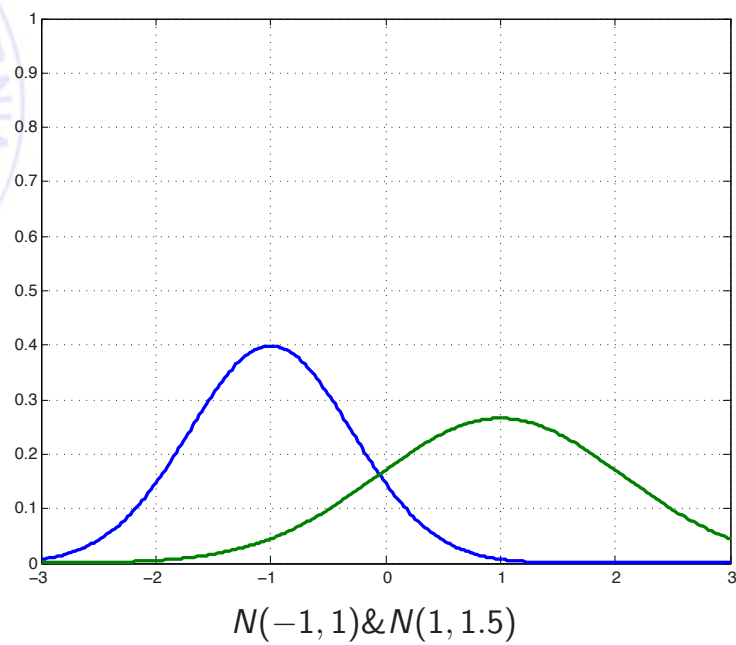
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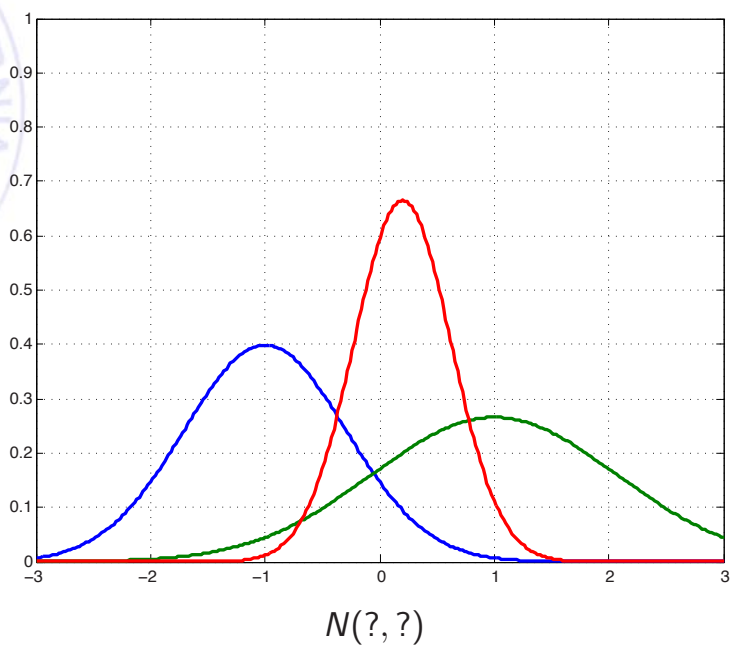
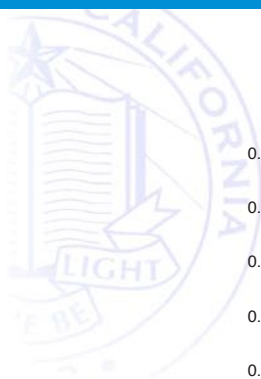
A small example - I



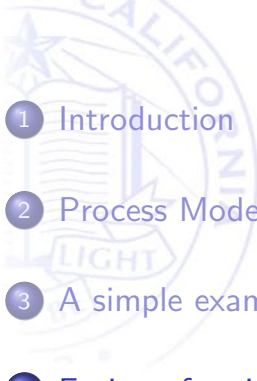
A small example - II



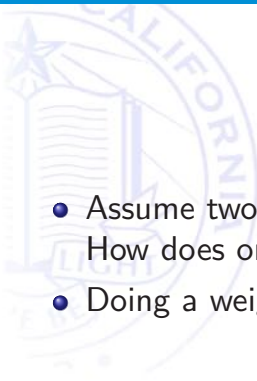
A small example - III - Joint value?



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Fusion of stochastic variables

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- Assume two measurement x_1 and x_2 with associated uncertainties σ_1 and σ_2 . How does one generate an optimal estimate $\hat{x}$?
 - Doing a weighted least square:

$$S = \sum_{i=1,2} w_i (\hat{x} - x_i)^2$$

what are the optimal weights w_i ?

- From $\frac{\partial S}{\partial \hat{x}} = 0$ we get ...

Fusion of stochastic variables



$$\hat{x} = \frac{\sum w_i x_i}{\sum w_i}$$

with $w_i = \frac{1}{\sigma_i^2}$ we get

$$\hat{x} = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2} x_1 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} x_2$$

and

$$\sigma_{\hat{x}} = \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2}$$

Weighted updating

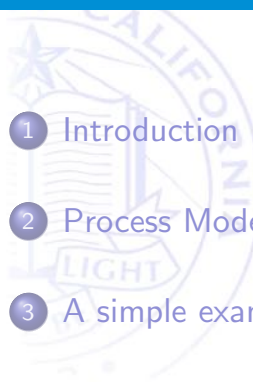


- The update can be rewritten to

$$\hat{x} = x_1 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} (x_2 - x_1)$$

- I.e. the updating = the value + a correction term

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The Kalman model

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- Composed of a prediction and an update of the estimate
 - Prediction is an estimate of what the system ought to be given our history / state.
 - Update is based on the difference between expectation and actual measurements.

Kalman Prediction



$$\begin{aligned}s_{t|t-1} &= Fs_{t-1|t-1} + Gu_t \\ \Sigma_{t|t-1} &= F\Sigma_{t-1|t-1}F^T + Q_t\end{aligned}$$

where Q is the uncertainty of the model / system noise

Kalman Updating



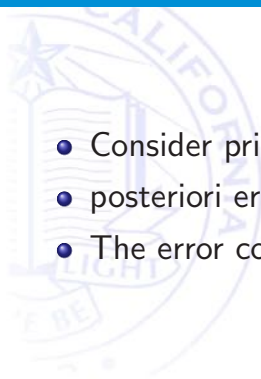
$$\begin{aligned}s_{t|t} &= s_{t|t-1} + K_t(z_t - Hs_{t|t-1}) \\ K_t &= \Sigma_{t|t-1}H^TS_t^{-1} \\ S_t &= H\Sigma_{t|t-1}H^T + R_t \\ \Sigma_{t|t} &= (I - K_tH)\Sigma_{t|t-1}\end{aligned}$$

Kalman Updating



$$\begin{aligned}s_{t|t} &= s_{t|t-1} + K_t(z_t - Hs_{t|t-1}) \\ K_t &= \Sigma_{t|t-1}H^T S_t^{-1} \\ S_t &= H\Sigma_{t|t-1}H^T + R_t \\ \Sigma_{t|t} &= (I - K_tH)\Sigma_{t|t-1}\end{aligned}$$

A bit of analysis



- Consider prior estimation error: $e_{t|t-1} = s_t - s_{t|t-1}$ and
- posteriori error $e_{t|t} = s_t - s_{t|t}$
- The error covariances are them:

$$\begin{aligned}\Sigma_{t|t-1} &= E[e_{t|t-1}e_{t|t-1}^T] \\ \Sigma_{t|t} &= E[e_{t|t}e_{t|t}^T] \\ K_t &= \Sigma_{t|t-1}H^T(\Sigma_{t|t-1}H^T + R_t)^{-1}\end{aligned}$$

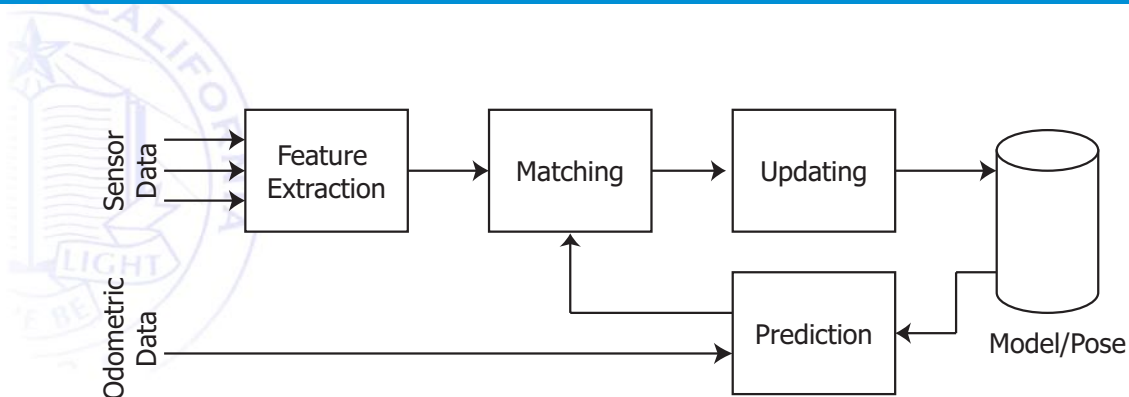
$$\lim_{R \rightarrow 0} K_t = H^{-1}$$

$$\lim_{\sigma_{t|t-1} \rightarrow 0} K_t = 0$$

Kalman interpretation

- F is the autonomous evolution
- H is the measurement prediction $p(z|s)$ is a prediction of where features in the world are in the sensory frame
- Σ is the uncertainty in the pose/state estimate $s_{t|t}$.
- S_t is the uncertainty in the sensory measurements
- R_t is the sensor noise
- Q_t is the uncertainty in the system model. How good is the model?

Example use of the Kalman filter

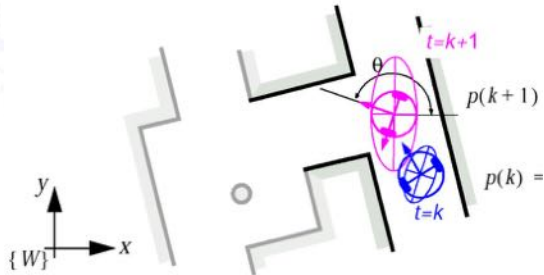


- Prediction / Update as described above
- Matching based on the Mahalanobis distance

$$M = z_t S_t^{-1} z_t^T$$

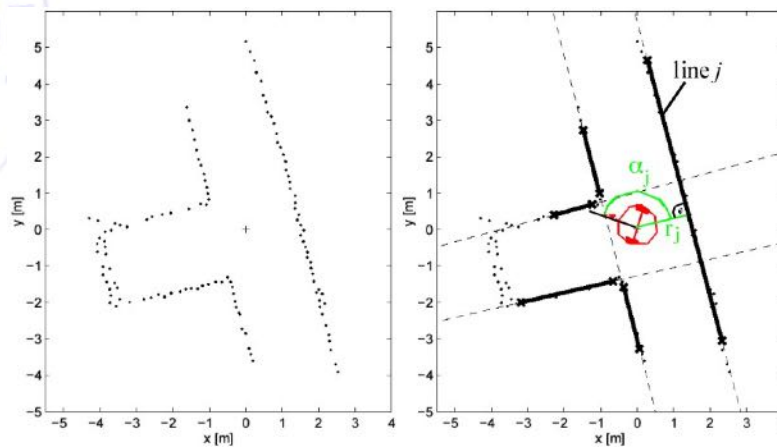
- A validation gate may be used: $M < \rho$

Kalman Example – Prediction

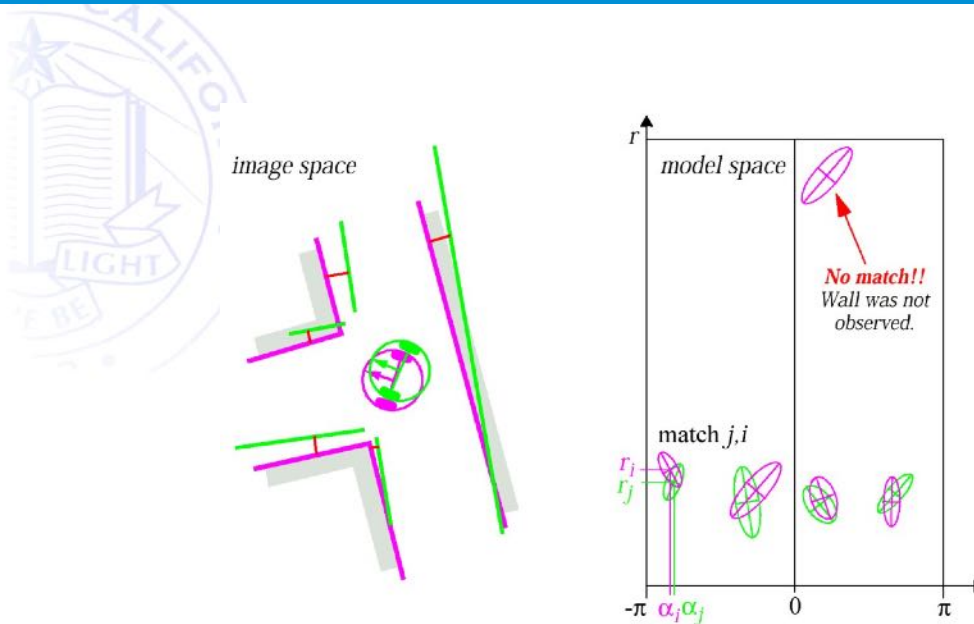


- Kinematic prediction of position and uncertainty
- Standard model

Kalman Example – Features

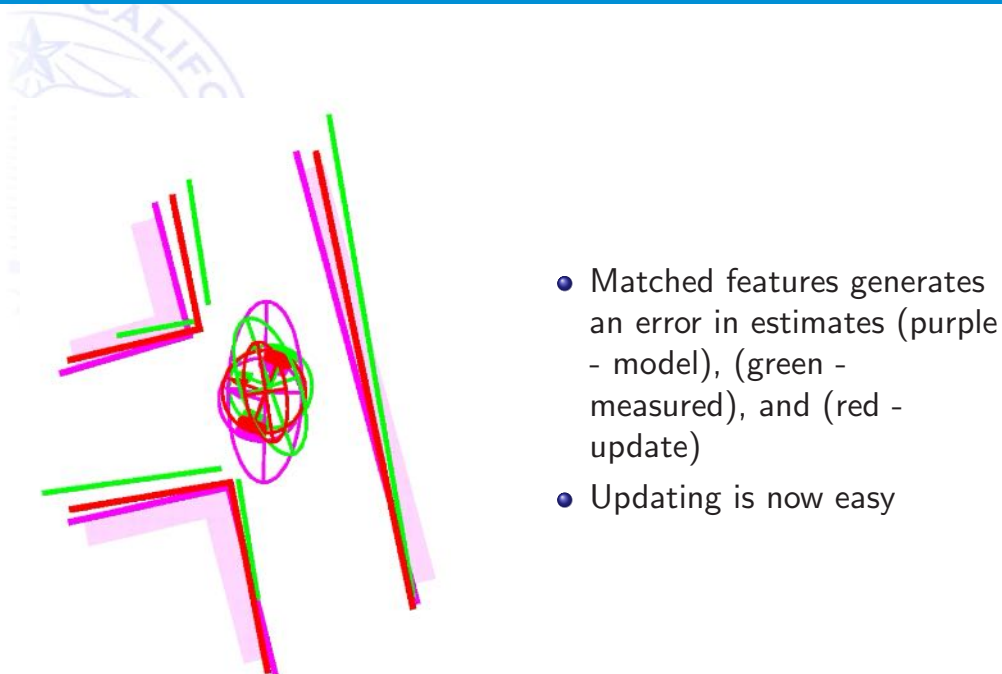


Kalman Example – Matching



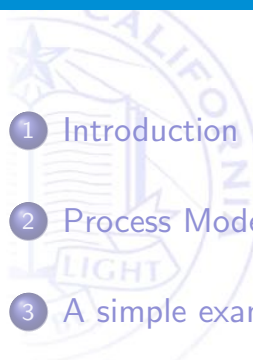
using nearest neighbour and predicted features matching is easy

Kalman Example – Updating

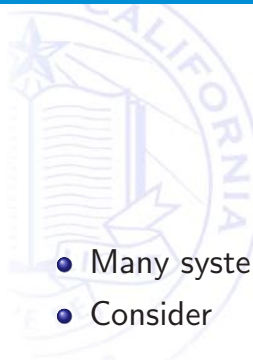


- Matched features generates an error in estimates (purple - model), (green - measured), and (red - update)
- Updating is now easy

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Non-linear systems

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- Many systems are non-linear, in these cases a linearization can be used
 - Consider

$$\begin{aligned}s_t &= f(s_{t-1}, u_{t-1}, w_{t-1}) \\ z_t &= h(s_t, v_k)\end{aligned}$$

A linearization of the system


$$\begin{aligned}s_{t|t} &\approx f(s_{t-1|t-1}, u_{t-1}, 0) + F(s_{t-1} - s_{t-1|t-1}) + Ww_{t-1} \\ z_t &\approx h(s_{t|t}, 0) + H(s_t - s_{t|t}) + Vv_k\end{aligned}$$

where

$$\begin{aligned}F_{ij} &= \frac{\partial f_i}{\partial s_j}(s_{t-1|t-1}, u_{t-1}, 0) \\ W_{ij} &= \frac{\partial f_i}{\partial w_j}(s_{t-1|t-1}, u_{t-1}, 0) \\ H_{ij} &= \frac{\partial h_i}{\partial s_j}(s_{t|t}, 0) \\ V_{ij} &= \frac{\partial h_i}{\partial v_j}(s_{t|t}, 0)\end{aligned}$$

The EKF computation



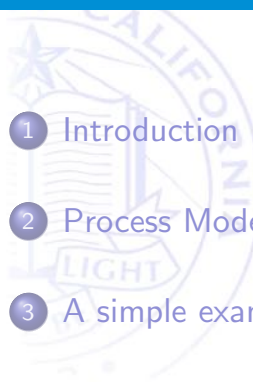
Time updating / prediction:

$$\begin{aligned}s_{t|t-1} &= f(s_{t-1|t-1}, u_{t-1}, 0) \\ \Sigma_{t|t-1} &= F_t \Sigma_{t-1|t-1} F_t^T + W_t Q_{t-1} W_t^T\end{aligned}$$

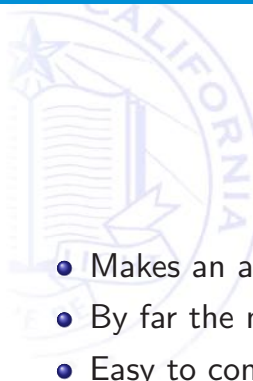
Measurement update computation

$$\begin{aligned}K_t &= \Sigma_{t|t-1} H_t^T (H_t \Sigma_{t|t-1} H_t^T + V_t R_t V_t)^{-1} \\ s_{t|t} &= s_{t|t-1} + K_t (z_t - h(s_{t|t-1}, 0)) \\ \Sigma_{t|t} &= (I - K_t H_t) \Sigma_{t|t-1}\end{aligned}$$

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Kalman – Discussion

- 
- Makes an assumption of a single pose estimate
 - By far the most frequently used model
 - Easy to compute.
 - Estimation of F and H can be difficult. For non-linear systems that are Jacobians that must be computed for each step.